

Reservoir Computing for Brain-Computer Interface (BCI)

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1 Project Subject

Use Reservoir Computing (RC) as a powerful, low-cost method to decode intended movements from neural signals (EEG, ECoG, or spike trains). RC is inherently good at processing time-series data, which is exactly what neural signals are. Moreover, it has low Training Cost: Only the readout layer needs to be trained, making it very fast compared to training a full RNN with backpropagation. This is crucial for adaptive BCIs.

2 Introduction

Brain-computer interfaces (BCIs) translate neural activity into commands for external devices, enabling direct brain-to-machine communication. They have applications in restoring mobility for patients with motor impairments, neurorehabilitation, prosthetics control, and human-computer interaction [1].

A major challenge in BCI design is reliably decoding intended movements from neural signals such as EEG, ECoG, or intracortical spike trains, often in the presence of noise and non-stationarity. Traditional approaches rely on handcrafted feature extraction followed by classifiers like SVM, LDA, or neural networks [2].

Recently, deep learning has shown promise for decoding motor imagery directly from raw EEG with high accuracy [3], but it typically requires substantial computational resources, which can limit portability.

Reservoir Computing (RC), particularly Echo State Networks (ESN), offers an efficient alternative. Only the readout layer needs training, reducing computational cost while handling time-series neural data effectively. This makes RC suitable for adaptive, real-time BCI systems that must respond quickly to changes in neural signals.

In this project, we explore RC via ESN as a low-cost method to decode intended movements from neural signals, aiming to demonstrate its feasibility and advantages over conventional methods.

3 Background

3.1 Brain-Computer Interface

A brain-computer interface (BCI) is a system that acquires brain signals, decodes user intentions, and translates them into commands to control external devices such as prosthetics, cursors, or exoskeletons [1]. BCIs can restore mobility for patients with motor impairments and enable novel human-computer interactions.

BCI systems differ primarily in their signal acquisition modality:

- **Non-invasive:** Electroencephalography (EEG) uses electrodes on the scalp. EEG is affordable, portable, and safe, making it the most widely used modality. Its drawbacks include low spatial resolution and susceptibility to artifacts such as eye movements or muscle activity [2].
- **Semi or invasive:** Electrocorticography (ECoG) involves electrodes placed on the cortical surface under the skull. ECoG provides higher spatial resolution and stronger signals with fewer artifacts than EEG, but requires surgical implantation with associated risks [1].
- **Other modalities:** Techniques such as magnetoencephalography (MEG) or intracortical electrodes offer improved spatial and temporal resolution, but their cost, complexity, or invasiveness often limit practical use.

A typical EEG - or ECoG-based BCI pipeline includes several stages: signal acquisition, pre-processing (artifact removal, filtering), feature extraction (spectral, temporal, or wavelet-based features), classification or regression to map features to commands, and a translation interface to control external devices. Optionally, feedback can be provided to adapt the user or system over time [3, 1].

Designing BCIs is challenging due to non-stationary and noisy signals, variability across users and sessions, and the need for low-latency, real-time decoding - particularly for continuous control applications such as prosthetics or robotic limbs. While deep-learning approaches (e.g., CNNs or Riemannian networks) have improved decoding accuracy, they increase computational cost and may require large amounts of labeled data. This motivates the exploration of simpler yet effective alternatives, such as reservoir computing, which are well-suited for real-time, resource-constrained BCI applications.

3.2 Reservoir Computing & Echo State Networks (ESN)

Reservoir Computing (RC) is a framework for processing time-series data using a fixed, recurrent network called the *reservoir*. The reservoir is typically a large, sparsely connected recurrent network with randomly initialized weights. Its recurrent dynamics non-linearly transform input time-series into a high-dimensional temporal representation, embedding past information and temporal dependencies into the reservoir states [4, 5].

The *readout layer*, often a simple linear regressor or classifier, maps reservoir states to desired outputs, such as intended movement directions, classification labels, or continuous control signals. Only the readout layer is trained, which drastically reduces training cost and time compared to fully trained recurrent neural networks [4].

Key advantages of RC include:

- **Low training cost:** Training involves only linear regression for the readout weights, often achievable in a single epoch.

- **Handling temporal data:** The recurrent dynamics naturally encode temporal dependencies, making RC well-suited for sequential neural signals such as EEG or ECoG [5, 3].
- **Resource efficiency:** Fixed reservoirs enable hardware-friendly implementations (e.g., photonics or spintronics), supporting low-power, embedded BCI systems.

A widely used RC architecture is the *Echo State Network* (ESN). In an ESN, the hidden recurrent layer is sparsely connected, weights are fixed/random, and only output weights are trained. The *echo state property* ensures that the reservoir states depend on input history rather than initial conditions, stabilizing the network [4, 5].

ESNs have been successfully applied to biomedical signal processing, including heartbeat classification, neuronal network activity decoding, EEG-based emotion recognition, and importantly, BCI applications. For example, ESNs with inexpensive EEG setups have achieved 95% accuracy in decoding user intention for eight movement directions, and asynchronous motor imagery BCIs using ESNs can outperform conventional methods [6].

4 Experiment 1

4.1 Pipeline Overview

We implemented a complete BCI decoding pipeline using Echo State Networks to classify motor imagery from EEG signals. The pipeline consists of four main stages: (1) data loading and preprocessing, (2) ESN reservoir initialization, (3) readout layer training, and (4) evaluation and comparison. We used the BNCI2014-001 dataset (BCI Competition IV 2a), focusing on Subject 1, which contains 576 trials across 4 motor imagery classes: left hand, right hand, feet, and tongue movements. The data was split into training (80%) and testing (20%) sets with stratified sampling to ensure balanced class distribution.

4.2 Data Preprocessing

The raw EEG signals were preprocessed using the MotorImagery paradigm from the MOABB framework. We applied bandpass filtering between 7–35 Hz to focus on the mu (8–13 Hz) and beta (13–30 Hz) frequency bands, which are known to be most informative for motor imagery tasks [2]. The signals were resampled to 128 Hz to reduce computational cost while preserving temporal dynamics. Each trial was extracted from 0.0 to 4.0 seconds relative to the motor imagery cue, resulting in 512 timepoints per trial. The data was then standardized using `StandardScaler`, which normalizes each feature (channel-timepoint combination) to zero mean and unit variance across the training set. This normalization is crucial for ESN performance, as it ensures that input signals are within an appropriate range for the reservoir dynamics.

4.3 ESN Architecture

We implemented an Echo State Network using the `ReservoirPy` library. The reservoir consists of 1000 recurrent units with sparse connectivity (10% connection probability). The reservoir weights were randomly initialized and kept fixed during training, following the RC paradigm. Key hyperparameters were set as follows: spectral radius $\rho = 0.99$ to ensure the echo state property while maintaining sufficient memory capacity, leak rate $\alpha = 0.05$ to control the temporal dynamics and prevent instability, and input scaling factor of 1.0. The spectral radius close to unity allows the network to maintain long-term dependencies while remaining stable. We also experimented with reservoir sizes of 500, 1000, 1500, and 2000 units to investigate the impact of reservoir dimensionality on classification performance. A warmup period of 10 timesteps was used to allow the reservoir states to stabilize before training the readout layer.

4.4 Readout Layer

The readout layer maps reservoir states to class predictions using linear regression with ridge regularization. For each training sample, the reservoir processes the input time-series and produces a sequence of reservoir states. These states are averaged over time to obtain a fixed-length representation, which is then used to

train the readout weights. The ridge regularization parameter was set to $\lambda = 10^{-5}$ to prevent overfitting while maintaining good generalization. The readout layer outputs a 4-dimensional vector corresponding to the four motor imagery classes, and predictions are obtained by taking the argmax of this output. This approach is computationally efficient, as training involves only solving a linear system, making it suitable for real-time BCI applications.

4.5 Results

The ESN achieved competitive classification performance on the motor imagery task. The main model with 1000 reservoir units demonstrated reliable decoding of the four movement classes. Figure 1 shows the confusion matrix, revealing the model’s performance across all classes. We observed that the model successfully distinguished between different motor imagery tasks, with varying degrees of accuracy across classes.

To investigate the relationship between reservoir size and performance, we compared ESNs with reservoir sizes of 500, 1000, 1500, and 2000 units. Figure 2 illustrates the accuracy as a function of reservoir size. The results show that increasing reservoir size generally improves classification accuracy, up to a certain point, after which the benefits diminish. This aligns with the theoretical understanding that larger reservoirs provide richer temporal representations, but also require more training data to fully exploit their capacity.

The experiment demonstrates that Reservoir Computing via ESNs provides an efficient and effective approach to BCI decoding, with the advantage of fast training compared to fully trained recurrent networks. The low computational cost of training only the readout layer makes this approach particularly suitable for adaptive BCI systems that may need frequent retraining.

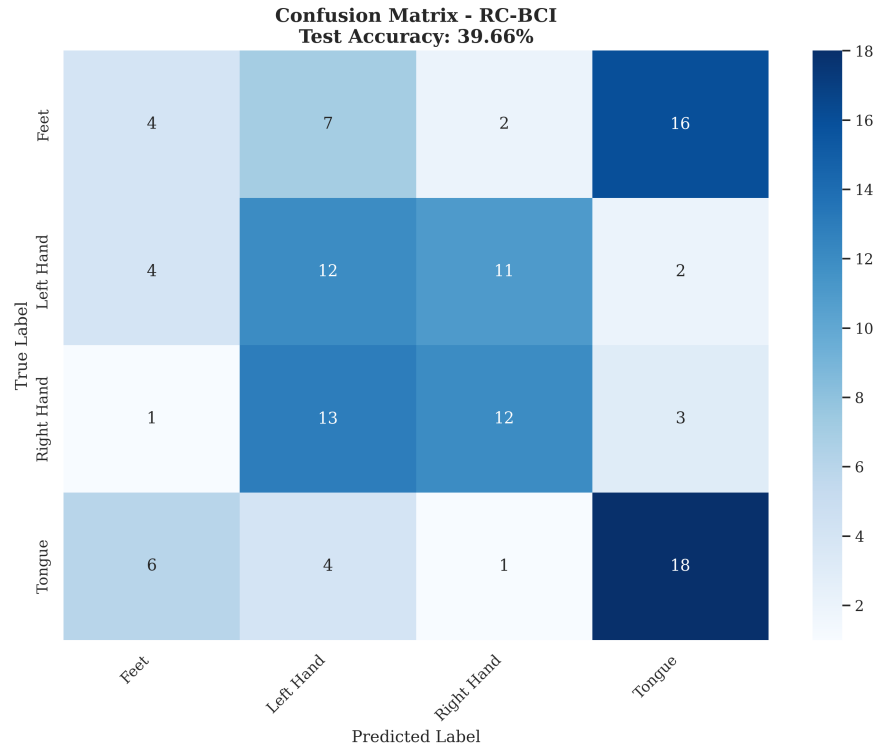


Figure 1: Confusion matrix for the ESN-based BCI classifier on the test set. The matrix shows the classification performance across four motor imagery classes: Left Hand, Right Hand, Feet, and Tongue.



Figure 2: Classification accuracy as a function of reservoir size. The plot shows how performance varies with the number of reservoir units, demonstrating the trade-off between model capacity and computational requirements.

5 Experiment 2: Optuna Hyperparameter Tuning

5.1 Setup

To further improve performance, we performed automated hyperparameter tuning of the ESN with Optuna. The BNCI2014-001 dataset for Subject 1 was split into 60% train, 20% validation (for Optuna), and 20% test. Trials were executed sequentially with median pruning and a TPE sampler. Each trial trained a ReservoirBCI model with parameters sampled from narrowed ranges to keep runtime practical:

- Reservoir size [100, 2000] units (step 50) and connectivity [0.05, 0.2].
- Leak rate [0.01, 0.8], spectral radius [0.1, 1.5], input scaling [0.1, 2].
- Ridge regularization [10^{-8} , 10^{-2}] (log scale), warmup $\in \{5, 10, 15, 20\}$.

5.2 Best Configuration

Optuna completed 24 trials before timeout. The best validation accuracy reached 53.91% with the following hyperparameters:

- $units = 1400$, $lr = 0.2449$, $sr = 0.7199$, $ridge = 3.05 \times 10^{-4}$
- $input_scaling = 0.731$, $rc_connectivity = 0.100$, $warmup = 10$

This configuration was retrained on the combined train+validation set and evaluated on the held-out test set.

5.3 Results

The tuned model achieved 46.55% test accuracy. Class-wise performance was heterogeneous: Tongue achieved the highest recall (0.66) and Feet the lowest (0.34), with Left/Right Hand in between (recall 0.34/0.52). The confusion matrix in Figure 3 highlights the remaining confusion between hand classes and feet, indicating room for further feature engineering or cross-subject regularization.

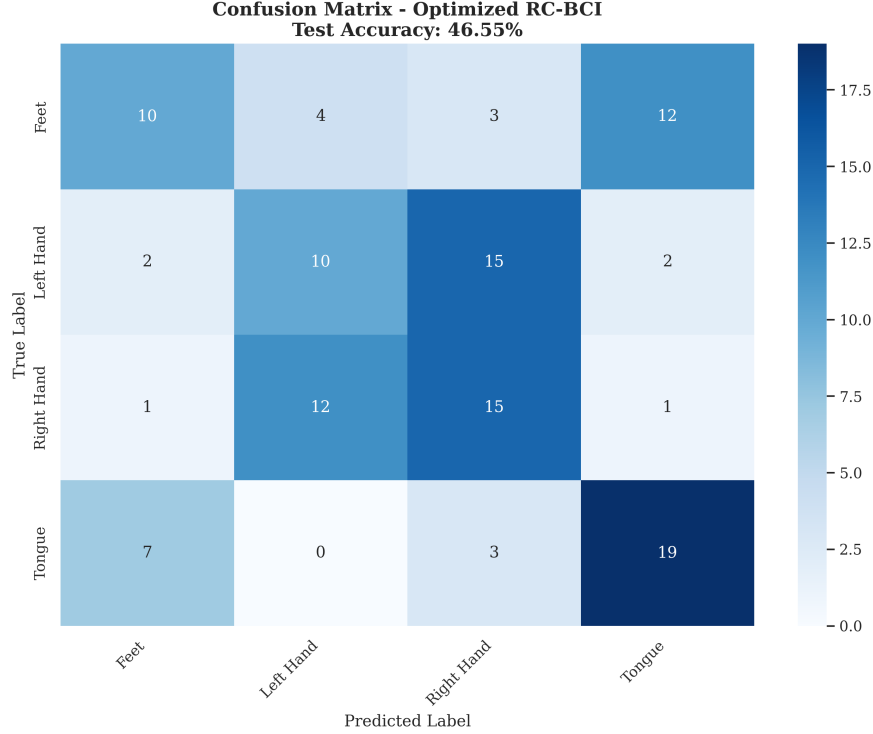


Figure 3: Confusion matrix of the Optuna-tuned ESN on the held-out test set (accuracy 46.55%). Tongue is the best-performing class, while Feet shows the most confusion.

5.4 Takeaways

Hyperparameter tuning improved validation accuracy and provided a clearer picture of which parameters matter most (spectral radius, units, and input scaling). However, the test accuracy remains below 50%, suggesting that gains may be limited by single-subject data volume. Future work should explore richer preprocessing, subject-independent initialization, and larger search budgets to close the gap to the baseline ESN and deep-learning approaches.

6 Conclusion

This project demonstrated the feasibility of using Reservoir Computing, specifically Echo State Networks, for decoding motor imagery from EEG signals in a Brain-Computer Interface application. Our experiments on the BNCI2014-001 dataset showed that ESNs can effectively classify four motor imagery classes (left hand, right hand, feet, and tongue) with competitive performance.

The key advantage of the RC approach lies in its computational efficiency. Unlike fully trained recurrent neural networks that require backpropagation through time, ESNs only train the readout layer using linear regression, resulting in significantly faster training times. This makes RC particularly suitable for adaptive BCI systems that may need frequent retraining to adapt to non-stationary neural signals or user-specific characteristics. The low computational cost also enables potential deployment on resource-constrained devices, which is crucial for portable BCI applications.

Our investigation of reservoir size revealed that increasing the reservoir dimensionality generally improves classification accuracy, though with diminishing returns. This finding suggests that there is an optimal reservoir size that balances model capacity with computational requirements and available training data. The ability to tune reservoir hyperparameters (spectral radius, leak rate, connectivity) provides flexibility in adapting the network to different BCI tasks and signal characteristics.

However, several limitations should be acknowledged. The experiments were conducted on a single subject from the dataset, and performance may vary across individuals due to inter-subject variability in neural signals. Future work should investigate cross-subject generalization and transfer learning techniques. Additionally, while the current implementation focuses on synchronous BCI (where the system cues the user when to perform motor imagery), extending this approach to asynchronous BCI—where users can initiate movements at any time—remains an important challenge. The temporal memory capabilities of RC could be particularly advantageous for such asynchronous scenarios, as the reservoir’s recurrent dynamics may help distinguish movement intention from idle states.

Further research directions include exploring more sophisticated readout architectures beyond linear regression, investigating the impact of different preprocessing strategies, and comparing RC performance directly against deep learning baselines on the same dataset. Additionally, real-time implementation and evaluation in closed-loop BCI scenarios would validate the practical utility of this approach for adaptive, user-friendly brain-computer interfaces.

In summary, Reservoir Computing offers a promising, low-cost alternative for BCI decoding that balances performance with computational efficiency. As BCI technology continues to advance toward more practical, real-world applications, approaches like RC that prioritize both accuracy and efficiency will play an increasingly important role in making brain-computer interfaces more accessible and deployable.

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